

Difference-in-Differences: The Library

Example

Experimentation, A/B Testing and Causal Inference (SDA-DSC-213), SDAIA Academy. Handout for Day 3, Hour 3 (slides 102 to 108). All five Day 3 handouts use the same story: a tutoring programme at one school.

Step 1. The story

School A opened a **study library** in 2025. School B did not. We have the pass rate of both schools before and after. Nobody was randomized; the library went to School A because its head teacher asked first.

Estimand **estimand: ATT on the pass rate**: the effect of the library on School A's 2025 pass rate, compared with what A's pass rate would have been without it.

Step 2. The 2 by 2 table (slide 103)

Pass rate (percent)	2024 (before)	2025 (after)	Change
School A (library)	60	72	+12
School B (no library)	50	56	+6

A's pass rate rose 12 points. But B's rose 6 points with no library: an easier exam, a better cohort, something common to both schools. The before/after change in A alone would overstate the library.

Step 3. The double difference

$$\tau_{\text{DiD}} = (Y_{A,\text{post}} - Y_{A,\text{pre}}) - (Y_{B,\text{post}} - Y_{B,\text{pre}}) = (72 - 60) - (56 - 50) = 12 - 6 = +6$$

The estimate **estimate: +6 points**. The first difference removes everything fixed about each school (A was always 10 points above B). The second removes everything that changed for both schools at once (the +6 common trend). What is left is attributed to the library.

Step 4. The same thing as a regression

Two-way fixed effects: a dummy for the school, a dummy for the year, and their interaction. The interaction coefficient is the DiD estimate.

$$Y = a + b \times \text{TreatedSchool} + c \times \text{Post} + \tau \times (\text{TreatedSchool} \times \text{Post}) + \text{error}$$

$$a = 50 \text{ (B before)} \quad b = 10 \text{ (A above B)} \quad c = 6 \text{ (common change)} \quad \tau = 6$$

With more schools and years this is the regression Lab 3 runs, with standard errors clustered by school because the treatment is assigned at the school level.

Step 5. The load-bearing assumption: parallel trends (slide 104)

The +6 is right only if, without the library, A's pass rate would have moved by the same +6 as B's. That is a statement about a world we cannot see. What we can check is whether the two schools moved in parallel **before** 2025:

Pass rate (percent)	2023	2024	Change 2023 to 2024
School A	56	60	+4
School B	46	50	+4

Parallel before. That supports the assumption; it does not prove it. A shock that hit only School A in 2025 (a new head teacher, a building closure) would still break it.

Step 6. Two diagnostics

Placebo in time. Pretend the library opened in 2024 and use only the pre-period years:

$$\text{placebo } \tau^{\wedge} = (60 - 56) - (50 - 46) = 4 - 4 = 0$$

A zero placebo is what a valid design should give. A large placebo effect means the schools were already diverging.

Event study (slide 105). With several years, estimate one coefficient per year relative to the opening. Pre-opening coefficients near zero and a jump after is the signature of a credible effect. With many schools opening libraries in different years, do not use plain two-way fixed effects; use a staggered-adoption estimator (slide 106).

Step 7. Synthetic control in three lines (slide 107)

If only one school gets a library and there are twenty possible comparison schools, build a weighted blend of the twenty whose pre-2025 pass rates track School A as closely as possible. The 2025 gap between the real A and the synthetic A is the estimate. Judge it by placebos: apply the same method to each untreated school and see whether A's gap stands out.

Three things to remember

1. DiD = change in the treated minus change in the untreated; it removes fixed differences and common shocks.
2. Parallel pre-trends support the assumption; nothing proves it. State it as the risk.
3. Cluster standard errors at the level the treatment was assigned.